

Coupled Entry Dynamics and Polynomial Embedding

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A Coupled Entry Dynamics and Polynomial Embedding

A.1 Coupled 6-DOF Entry Dynamics

The entry vehicle state comprises the coupled translational, rotational, thermal, ablative, and modal subsystems. Position is carried in Cartesian Earth-centred Earth-fixed (ECEF) components and attitude as a quaternion from the ECEF frame to the body frame. No angle is a state, so the dynamics have no coordinate singularities anywhere on the globe; geocentric latitude and longitude are read out from the position after integration.

Frames.

- $\mathcal{E}$: Earth-centred Earth-fixed (ECEF), rotating with the constant planetary rate $\vec{\omega}_E = \omega_E \hat{z}_{\mathcal{E}}$.
- $\mathcal{B}$: body frame, fixed at the instantaneous center of mass.

The attitude quaternion $\vec{q} = (q_0, q_1, q_2, q_3)^\top$ carries $\mathcal{E} \rightarrow \mathcal{B}$, with direction-cosine matrix

$$\mathbf{R} \equiv \mathbf{R}_{B/E}(\vec{q}) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}, \quad (\text{A.1})$$

quadratic in $\vec{q}$.

Position Kinematics. With $\vec{r} = (r_1, r_2, r_3)^\top$ the ECEF position and $\vec{V} = (u, v, w)^\top$ the planet-relative velocity resolved in $\mathcal{B}$, the kinematic equation is:

$$\dot{\vec{r}} = \mathbf{R}^\top \vec{V}. \quad (\text{A.2})$$

Translational Dynamics. Newton's law holds in the inertial frame; transporting the planet-relative velocity through $\mathcal{E}$ into $\mathcal{B}$, with inertial body rates $\vec{\omega} = (p, q, r_b)^\top$, mass m , aerodynamic force $\vec{F}_a$, thrust $\vec{F}_t$, and the planetary rate resolved in the body frame $\vec{\omega}_E^{\mathcal{B}} = \mathbf{R} \omega_E \hat{z}_{\mathcal{E}}$, the translational dynamical equations are:

$$\dot{\vec{V}}|_{\mathcal{B}} = \frac{1}{m}(\vec{F}_a + \vec{F}_t) + \mathbf{R} \vec{g}(\vec{r}) - (\vec{\omega} + \vec{\omega}_E^{\mathcal{B}}) \times \vec{V} - \vec{\omega}_E^{\mathcal{B}} \times (\vec{\omega}_E^{\mathcal{B}} \times \mathbf{R} \vec{r}). \quad (\text{A.3})$$

Gravity, including the J_2 oblateness term, in ECEF components:

$$\vec{g}(\vec{r}) = -\frac{\mu}{\|\vec{r}\|^3} \left[\left(1 + \frac{3}{2} J_2 \frac{R_e^2}{\|\vec{r}\|^2} \left(1 - 5 \frac{r_3^2}{\|\vec{r}\|^2} \right) \right) \vec{r} + 3 J_2 \frac{R_e^2}{\|\vec{r}\|^2} r_3 \hat{z}_{\mathcal{E}} \right]. \quad (\text{A.4})$$

Remark A.1. This is the gradient of the J_2 potential $\frac{\mu}{\|\vec{r}\|} [1 - \frac{1}{2} J_2 (R_e / \|\vec{r}\|)^2 (3r_3^2 / \|\vec{r}\|^2 - 1)]$; its radial and northward components reproduce the spherical form, with $r_3 / \|\vec{r}\| = \sin \phi$.

Rotational Dynamics / Euler's Equations. With angular momentum $\vec{h} = \mathbf{I} \vec{\omega}$, combined aerodynamic / reaction control system (RCS) moments $\vec{M} = \vec{M}_a + \vec{M}_{\text{RCS}}$, and the inertia tensor $\mathbf{I} = \mathbf{I}(s, \eta)$ depending on recession depth s and modal amplitudes η (so that $\dot{\mathbf{I}} \vec{\omega}$ is the variable-inertia term), the rotational dynamical equations are:

$$\mathbf{I} \dot{\vec{\omega}} = \vec{M} - \vec{\omega} \times (\mathbf{I} \vec{\omega}) - \dot{\mathbf{I}} \vec{\omega}. \quad (\text{A.5})$$

Quaternion Kinematics. The quaternion is driven by the body rate *relative to* $\mathcal{E}$, $\vec{\omega}_{B/E} = \vec{\omega} - \vec{\omega}_E^{\mathcal{B}}$. For $\vec{a} = (a_1, a_2, a_3)^\top$,

$$\dot{\vec{q}} = \frac{1}{2} \Omega(\vec{\omega} - \mathbf{R} \omega_E \hat{z}_{\mathcal{E}}) \vec{q}, \quad \Omega(\vec{a}) = \begin{pmatrix} 0 & -a_1 & -a_2 & -a_3 \\ a_1 & 0 & a_3 & -a_2 \\ a_2 & -a_3 & 0 & a_1 \\ a_3 & a_2 & -a_1 & 0 \end{pmatrix}. \quad (\text{A.6})$$

Remark A.2. The rate is **polynomial** — bilinear in $(\vec{\omega}, \vec{q})$ and cubic in $\vec{q}$ through the Earth-rate term. With this Ω and $\mathbf{R}$ of Eq. (A.1), $\dot{\mathbf{R}} = -[\vec{\omega}_{B/E} \times] \mathbf{R}$ holds identically, and $\Omega(\vec{a})$ is skew for every $\vec{a}$, so $|\vec{q}|$ is preserved.

Thermal. With the density-square-root embedding $y = \sqrt{\rho/\rho_0}$ and $V = \sqrt{u^2 + v^2 + w^2}$, the modulus lift $\dot{\chi} = c \chi \dot{T}_w$ replaces T_w with a polynomial via Varma–Savageau–Voit recasting. Stagnation-point heating (Sutton–Graves, with nose radius R_n , sea-level density ρ_0) equations are then given by:

$$\dot{Q}_{\text{tot}} = \dot{Q} = k_Q \sqrt{\rho_0/R_n} y V^3, \quad \dot{T}_w = \frac{\dot{Q} - (T_w - T_\infty)/\tau_{\text{cond}}}{C_w}. \quad (\text{A.7})$$

Ablation. With recession rate coupled to heat flux and mass loss from char-layer blow-off, the ablative equations are given by:

$$\dot{s} = \alpha_s \dot{Q}, \quad \dot{m} = -\dot{m}_{\text{abl}}. \quad (\text{A.8})$$

Modal/Structural Elasticity. With piston-theory forcing and temperature-dependent natural frequency, per mode i our modal equations are:

$$\ddot{\eta}_i + 2\zeta_i \omega_i(T) \dot{\eta}_i + \omega_i(T)^2 \eta_i = Q_i(x, \eta, \dot{\eta}). \quad (\text{A.9})$$

A.2 Polynomial Embedding via Adjunction

A priori, the dynamics contain transcendental and rational terms. We repair each of these by adjoining one variable and one polynomial defining constraint, so the entire augmented system is polynomial in the augmented state:

1. **Transcendental Incidence:** $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin(v/V)$ are inverse-trigonometric functions of body rates, we replace them with polynomial direction-cosine ratios $(u/V, v/V, w/V)$.
2. **Square-Root Speed:** Since $V = \sqrt{u^2 + v^2 + w^2}$ carries a square root, we adjoin V with the velocity magnitude constraint $V^2 = u^2 + v^2 + w^2$.
3. **Exponential Atmosphere:** Since $\rho = \rho_0 e^{-h/H_s}$ with $h = \|\vec{r}\| - R_e$ appears in $\bar{q} = \frac{1}{2} \rho V^2$, we adjoin $y = \sqrt{\rho/\rho_0}$ with rate $\dot{y} = -\frac{y}{2H_s} \frac{d}{dt} \|\vec{r}\|$, where the radial rate $\frac{d}{dt} \|\vec{r}\| = \varrho(\mathbf{R}\vec{r}) \cdot \vec{V}$ is polynomial once ϱ (item 4) is adjoined.
4. **Reciprocal Radius:** Gravity carries $\|\vec{r}\|^{-3}$, $\|\vec{r}\|^{-5}$, $\|\vec{r}\|^{-7}$, so we adjoin $\varrho = 1/\|\vec{r}\|$ with $\varrho^2 \|\vec{r}\|^2 = 1$, $\varrho > 0$, and rate $\dot{\varrho} = -\varrho^3 (\mathbf{R}\vec{r}) \cdot \vec{V}$. Gravity is then polynomial:

$$\vec{g} = -\mu \varrho^3 \left[\left(1 + \frac{3}{2} J_2 R_e^2 \varrho^2 (1 - 5 r_3^2 \varrho^2) \right) \vec{r} + 3 J_2 R_e^2 \varrho^2 r_3 \hat{z}_{\mathcal{E}} \right]. \quad (\text{A.10})$$

No trigonometric adjunctions are needed: no angle is a state, and latitude dependence enters only through $r_3 \varrho = \sin \phi$.

5. **Quaternion Norm:** $|\vec{q}|^2 = 1$ is already a polynomial invariant (skew-symmetry of Ω preserves it), no adjunction necessary.
6. **Heat / Load-Factor Square Roots:** $\dot{Q} \propto yV^3$ is already polynomial; however, $n = \|\vec{F}_a\|/(mg_0)$ requires squaring to $(\tilde{q}S)^2(C_X^2 + C_Y^2 + C_Z^2) \leq (n_{\max}mg_0)^2$.

The Augmented State. Collecting all coordinates:

$$\tilde{x} = \left(\underbrace{r_1, r_2, r_3, u, v, w, q_0, q_1, q_2, q_3, p, q, r_b}_{13, \text{ core}}, \underbrace{y, V, \varrho, Q_{\text{tot}}, m, s, \chi}_{7, \text{ adjoined}}, \underbrace{\eta, \dot{\eta}}_{2n_\eta, \text{ modal}} \right). \quad (\text{A.11})$$

References